Equation 15: the concrete substitution map

The upper line has already combined the lower cases. There is no remaining $\mathrm{d}x_\ell/\mathrm{d}\alpha$ slot in it. To see exactly how they connect, take two uncapped lanes, one capped lane, and an optional fourth lane that remains at zero:

$$\mathcal{A} = \{1, 2\}, \qquad \mathcal{U} = \{3\}, \qquad x_4 = 0.$$

First map the categories to the upper formula.

Upper-formula slot	Replacement	Reason
$\lvert\mathcal{A}\rvert$	2	Lanes 1 and 2 are uncapped.
$\displaystyle\sum_{\mathcal{U}} \frac{\mathrm{d}u_\ell}{\mathrm{d}\alpha}$	$\displaystyle\frac{\mathrm{d}u_3}{\mathrm{d}\alpha}$	Only lane 3 is capped.
$\displaystyle\sum_{\mathcal{A}} \frac{\mathrm{d}h_\ell}{\mathrm{d}\alpha}$	$\displaystyle\frac{\mathrm{d}h_1}{\mathrm{d}\alpha} + \frac{\mathrm{d}h_2}{\mathrm{d}\alpha}$	Only lanes 1 and 2 enter this sum.

Thus the upper line becomes

$$\boxed{\frac{\mathrm{d}\lambda}{\mathrm{d}\alpha} = \frac{1}{2}\left(\frac{\mathrm{d}V}{\mathrm{d}\alpha} - \frac{\mathrm{d}u_3}{\mathrm{d}\alpha} + \frac{\mathrm{d}h_1}{\mathrm{d}\alpha} + \frac{\mathrm{d}h_2}{\mathrm{d}\alpha}\right).}$$

Here is precisely where the lower cases were substituted. The starting point is the derivative of $x_1 + x_2 + x_3 + x_4 = V$:

$$\begin{aligned}
\frac{\mathrm{d}V}{\mathrm{d}\alpha} &= \frac{\mathrm{d}x_1}{\mathrm{d}\alpha} + \frac{\mathrm{d}x_2}{\mathrm{d}\alpha} + \frac{\mathrm{d}x_3}{\mathrm{d}\alpha} + \frac{\mathrm{d}x_4}{\mathrm{d}\alpha} \\
&= \underbrace{\left(\frac{\mathrm{d}\lambda}{\mathrm{d}\alpha} - \frac{\mathrm{d}h_1}{\mathrm{d}\alpha}\right)}_{\text{lane 1: uncapped}} + \underbrace{\left(\frac{\mathrm{d}\lambda}{\mathrm{d}\alpha} - \frac{\mathrm{d}h_2}{\mathrm{d}\alpha}\right)}_{\text{lane 2: uncapped}} + \underbrace{\frac{\mathrm{d}u_3}{\mathrm{d}\alpha}}_{\text{lane 3: capped}} + \underbrace{0}_{\text{lane 4: zero}} \\
&= 2\frac{\mathrm{d}\lambda}{\mathrm{d}\alpha} - \frac{\mathrm{d}h_1}{\mathrm{d}\alpha} - \frac{\mathrm{d}h_2}{\mathrm{d}\alpha} + \frac{\mathrm{d}u_3}{\mathrm{d}\alpha}.
\end{aligned}$$

Rearranging this last line gives the boxed upper formula.

Now calculate: upper first, then lower. Suppose the known input sensitivities are

$$\frac{\mathrm{d}V}{\mathrm{d}\alpha} = 0.3, \quad \frac{\mathrm{d}u_3}{\mathrm{d}\alpha} = 0.1, \quad \frac{\mathrm{d}h_1}{\mathrm{d}\alpha} = 0.2, \quad \frac{\mathrm{d}h_2}{\mathrm{d}\alpha} = 0.$$

The upper formula gives $\mathrm{d}\lambda/\mathrm{d}\alpha = (0.3 - 0.1 + 0.2 + 0)/2 = 0.2$. Insert this result into each lower case:

$$\underbrace{\frac{\mathrm{d}x_1}{\mathrm{d}\alpha} = 0.2 - 0.2 = 0}_{\text{uncapped: level minus load}}, \quad \underbrace{\frac{\mathrm{d}x_2}{\mathrm{d}\alpha} = 0.2 - 0 = 0.2}_{\text{uncapped: level minus load}}, \quad \underbrace{\frac{\mathrm{d}x_3}{\mathrm{d}\alpha} = 0.1}_{\text{follows its cap}}, \quad \underbrace{\frac{\mathrm{d}x_4}{\mathrm{d}\alpha} = 0}_{\text{stays zero}}.$$

Check: the four individual gradients sum to $0 + 0.2 + 0.1 + 0 = 0.3$, exactly the given total-admission gradient. These are sensitivities, not mass values; the example assumes the lane categories remain unchanged.